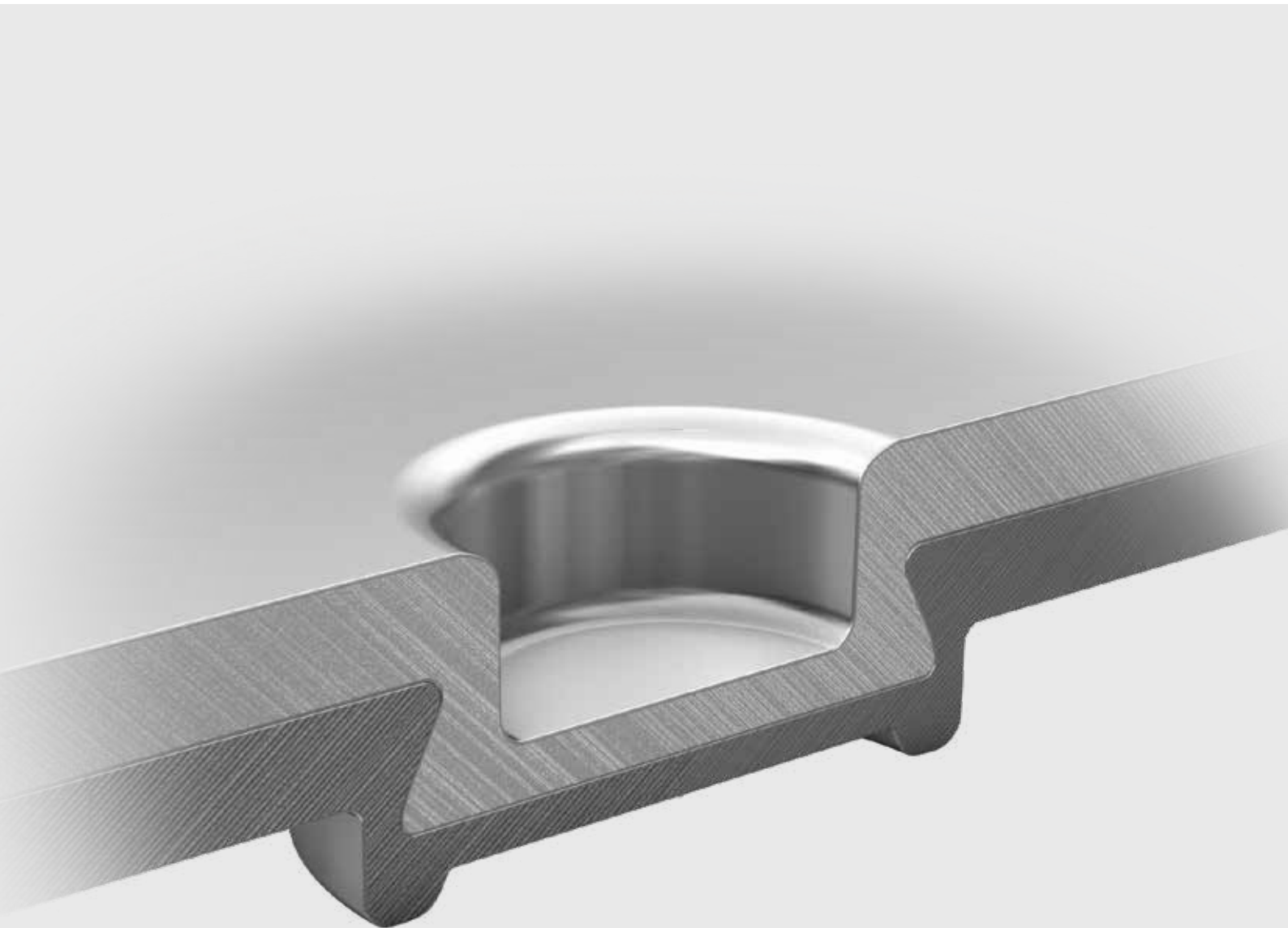


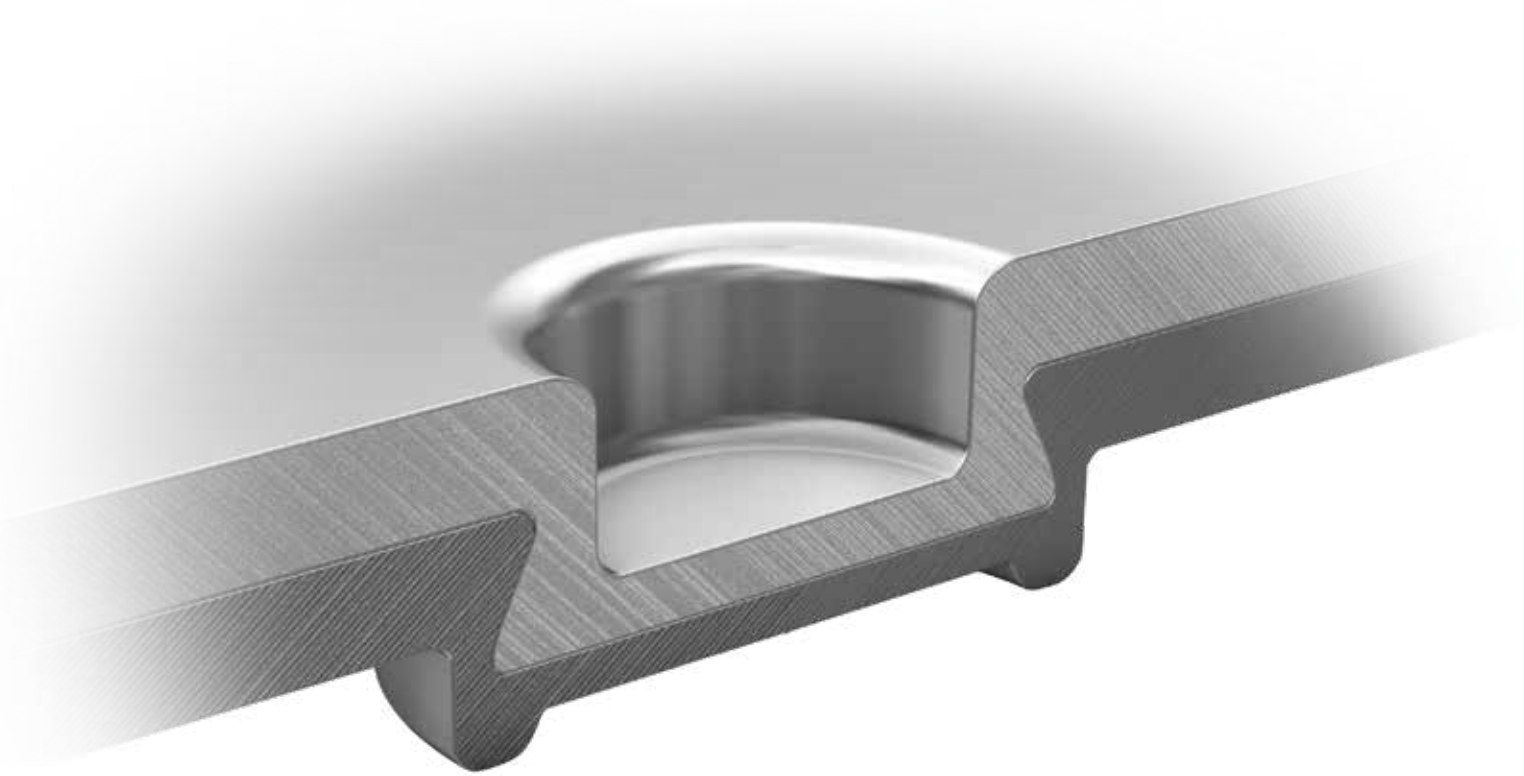


TOX[®] Clinching Technology

Riveting without rivets



Clinching - the high efficient joining technology



The TOX® Clinch Joint:
a buttonlike connection,
that holds sheet metal
together.

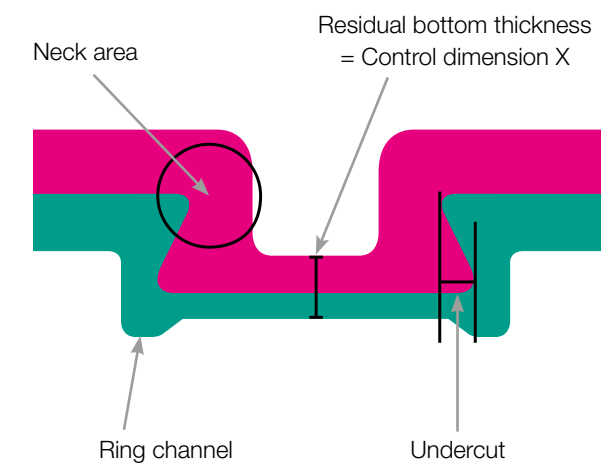
TOX® Clinching, clinch joining or „toxing“ is an excellent method for permanently joining sheet metal. It is based on efficient cold forming and requires no additional joining elements such as screws or rivets.

A punch and a die with a fixed or movable die are used for clinching. The technology offers a robust alternative to welding by using precise pressure to create permanently stable connections without negatively affecting the material properties. This makes clinching particularly suitable for joining sheets of different material thicknesses and types.

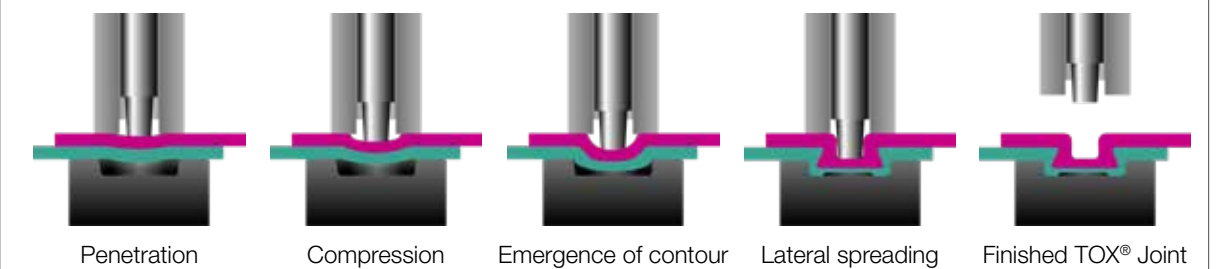
The high process reliability and cost-effectiveness make clinching a favoured choice in many industrial applications, as it eliminates the need for additional fasteners while ensuring a strong and durable joint.

Why does a TOX® Joint hold?

In the TOX® Clinching process a punch presses material into a die. Through plastic deformation and reverse flow of the material the joint forms an undercut to generate the holding force.



Creation of a TOX® Clinching Joint



The low-emission joining process versatile and cost-saving

Flexibility

With the clinching process, sheet metals of different qualities, surfaces and thicknesses can be joined, also in combination with adhesives or intermediate layers.

Different materials

TOX® Clinching allows a wide range of material combinations to be joint, including steel, aluminium, stainless steel and copper. Thanks to continuous further development of the tools, a long service life and a reliable connection are achieved.

Excellent cost savings

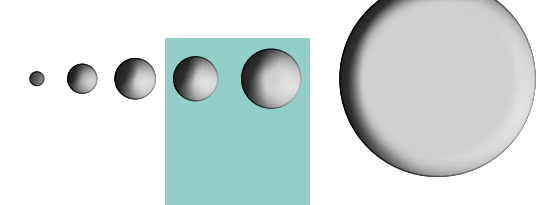
Directly compared to spot welding, TOX® Clinching is approximately 40 % cheaper with regard to investment, operating and tool costs. The clinching-tools have a long service life which reduces the investment. It offers low ongoing operating costs, as no additional materials or elements are necessary. Automation enables rational production with no post-processing and allows continuous quality control.

Advantages

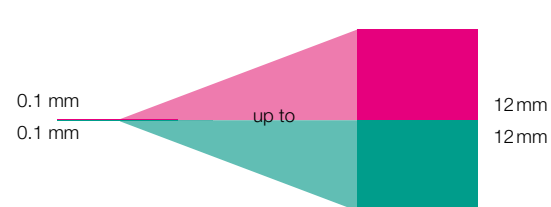
- Longlasting strong connections
- Allows wide range of material combinations
- Joining in combination with adhesives or intermediate layers
- Long service life of tools
- High degree of automation
- Continuous process- and quality monitoring

Point sizes:

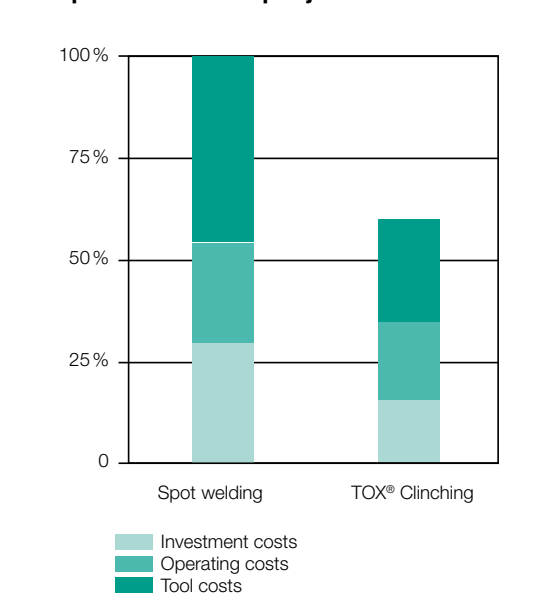
Diameter 2 – 26 mm



Sheet metal thicknesses



Comparison of costs per joint



Low-emission and energy-efficient

As clinching is a cold forming process, there are positive environmental aspects including:

- Low energy consumption because there is no heat input required
- No consumption of coolants or oils and no pollution of might harmful gases
- No disposal of chemicals or waste
- Low noise despite short cycle times

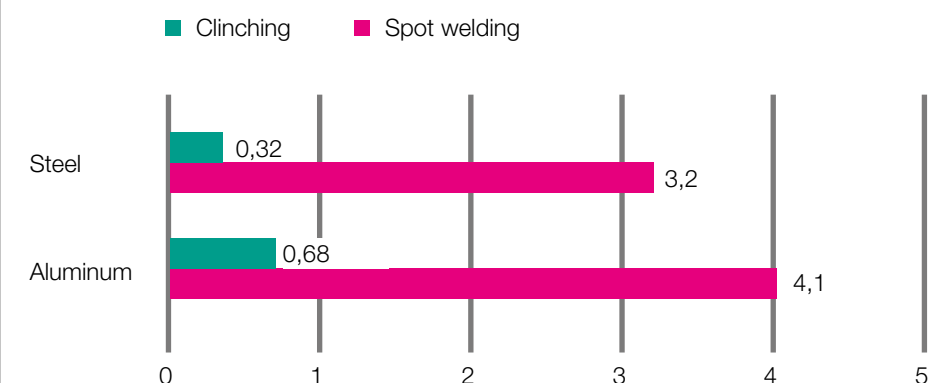


Clinching beats welding

Compared to welding, clinching offers clear advantages. All the tests carried out by the Laboratory for Materials and Joining Technology at Paderborn University speak in favor of the TOX® Technology. The connections are stronger and last longer and have a better electrical conductivity. Regarding investment and operating costs to sustainability and environmental aspects - clinching is leading.



Energy consumption per joint in watt hours (Wh)



Materials and combinations



Continuous proven joining quality

Strong and longlasting

High static strength

Clinch connections achieve high static retaining forces of up to 70% of a weld point. They have a positive strain hardening, maintain the surface coating and are generally more economical.

Corrosion-free joining technology

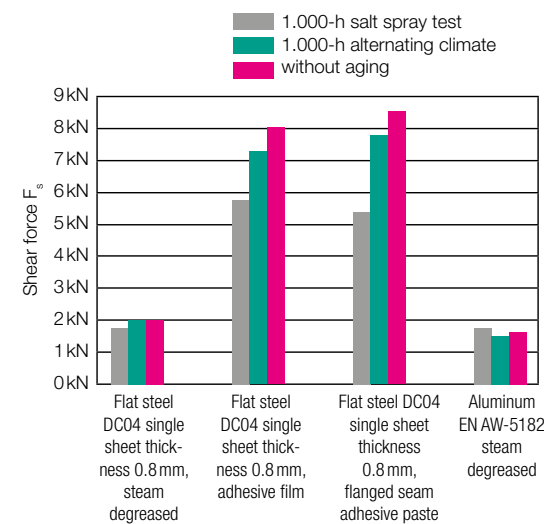
Because the surface coating is not damaged during clinching the anti-corrosive properties of the joined sheet metals (e.g. galvanization) are maintained.

Consistent dynamic strength

In comparison to spot welding, the clinching process does not have the negative properties of the notch effect. This allows the joint strength to remain constant over various load cycles.

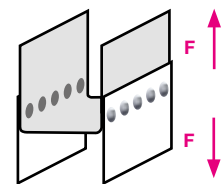
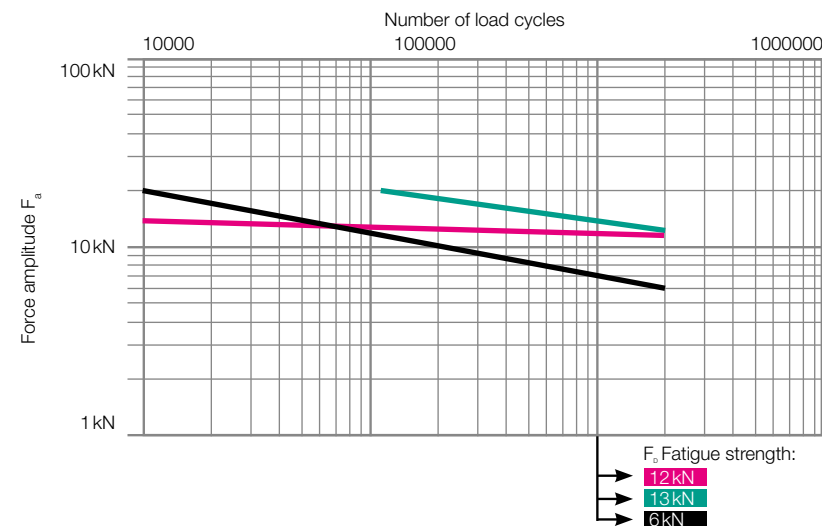
TOX® Clinching Joint Corrosion test

Holding strength with different materials and conditions



Dynamic strength

Strength of clinch, punch rivet and spot welding connections under oscillating shear strength load (endurance fatigue test).



Material:
Flat steel ZStE 420
Single sheet thickness 1 mm

Joining technologies:
Clinching TOX® Round-Joint
 $d_n = 8$ mm

Riveting with semi-tubular
rivets 5.3 x 5.0 mm

Spot welding
 $d_s = 5$ mm

Source: FOSTA P283

Testing in the TOX® Technical Center

Prior to collaboration TOX® works on the most effective solution for you in its laboratory. Here we will perform preliminary joining tests on your samples, which we test and analyze afterwards.

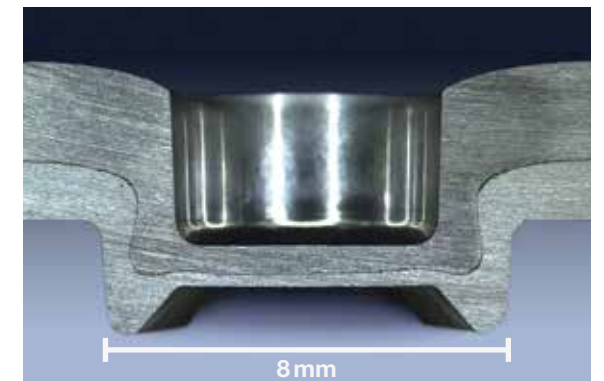
We will determine all parameters for your application, e.g. the required press force and suitable tools. Final test reports ensure the quality of the connection.

Quality monitoring during production

A significant advantage of clinching is the simple quality control in series production. By continuously measuring the residual bottom thickness each clinch joint can be checked. An additional analysis can be carried out using cross sections (cut through the point). The shear and pull strength can be determined via tensile tests.

Advantages

- High dynamic and static joint strength
- Proven joining quality in pre-test and then in series production
- No damage to coatings and surfaces
- Excellent electrical conductivity

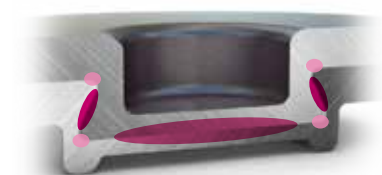


With a cross section of the clinching joint (cut through the point) the exact formation can be examined under a microscope. If necessary, optimizations can be made.

TOX® eClinching

The clinching process provides the best electrical conductivity compared to other mechanical joining processes such as soldering, welding and screw connections. During the eClinching process, a partial material bonding occurs in the joint zone in addition to the form and force fit: the joined sheets flow into each other in such a way that a large number of micro-contacts are created.

TOX® eClinching ensures maximum conductivity through precise connections of electrical components.



Area of a TOX® eClinching joint with the best material adhesion:

- very high (neck area)
- high (bottom)
- good

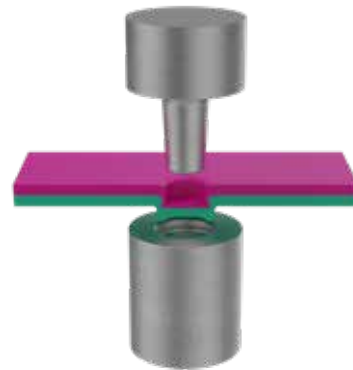


TOX® Clinching Technology: The perfect joint for every application

TOX® provides you with the best technological solutions and complete systems from one source, the respective design will be tailored to your specific joining tasks. An important step here is the selection of the clinch point shape:

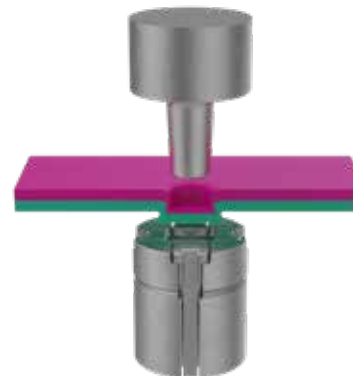
TOX® Round Joint

The classic: A round punch presses the materials to be joined into the die. The result is an aesthetically pleasing round button, which joins clearly without any burrs or sharp edges the sheet metals. The protective layer, e.g. paint, flows and does not get damaged.



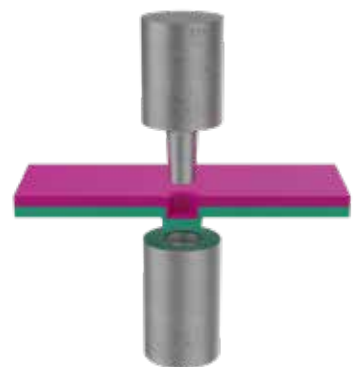
TOX® SKB

The special die consists of solid and movable segments. The solid segments center the material and the punch. The movable elements allow flow of the material in the joint – especially with sheet metal of different thickness or when using adhesives between the sheet layers.



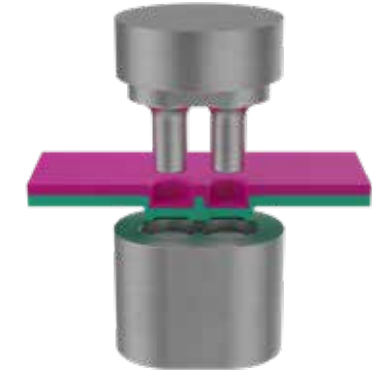
TOX® MICROpoint

The miniature TOX® Round-Joint is our answer to component miniaturization. With a diameter of 2mm, it is ideal for sheet metal thicknesses of 0.1–0.5mm and for slim flanges. For this reason, it is also optimally suited as an eClinch-Point.



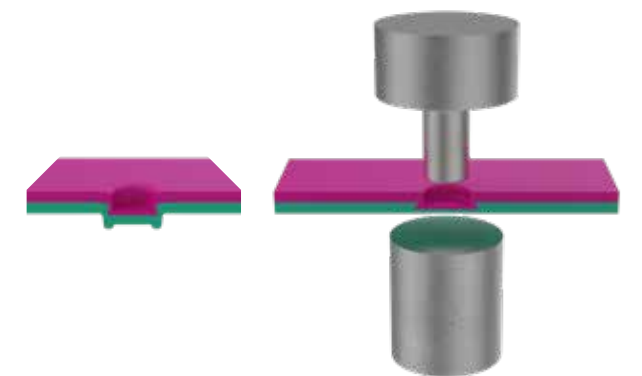
TOX® TWINpoint

This double cavity die provides protection against rotation and increases the joint strength compared to the single joint. It is ideal for small flanges and limited installation space. When eClinching, it also makes sure the design requirements are met.



TOX® Flat Point

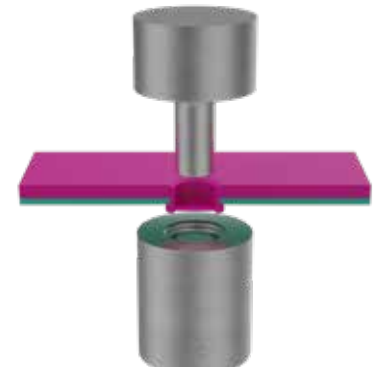
If the protrusion on die side presents a problem: with the TOX® Flat Point, the protrusion of the clinching joint on die side is pressed flat again in a second round using a flat die. A minimum protrusion height of only a few tenths of a millimeter remains.



TOX® Vario-Point

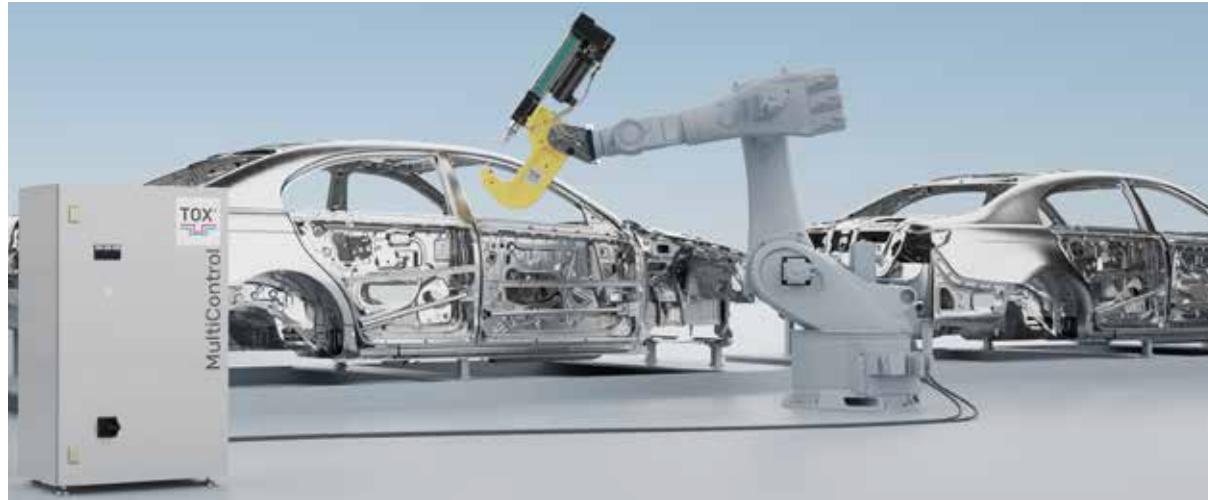
The TOX® Joint for difficult combinations: Joining of sheet metal with large differences in thickness, joining of high strength or non ductile materials with ductile materials or joining of non metallic materials.

The sheet metal is driven through the pre-punched material for this purpose. The process requires low joining forces and can also be applied in multi-point tools.



TOX® Multi Technology Platform

The industrial production system



Clinching, also called toxing, has become indispensable in automated manufacturing. The ever-changing companies of the automotive industry in particular rely on the cold joining process because a wide range of joining tasks can be carried out extremely quickly and without additional materials.

The reason for this is (energy) efficiency, on the one hand. On the other hand, it is electrical conductivity, which, due to the continuous development towards electric mobility, is more in demand than ever. With the TOX® Multi Technology Platform clinching truly comes into its own.

Advantages

Process times

Extremely short cycle times are required in many automated production lines and processes. These can be achieved quickly and reliably with the TOX® Clinching System. Through minimal process times, we increase your productivity - from day one.

Intelligent tongs: TOX® Clinching Frame

Tong-specific data such as calibration factors, maintenance counters and maximum values are not stored on the controller, but on a retentive memory on the tong. This means that you never lose track of the data, even when changing a frame.

The pneumatic TOX® ToolCheck provides additional intelligence on the tongs. This checks the tools and detects possible defects.

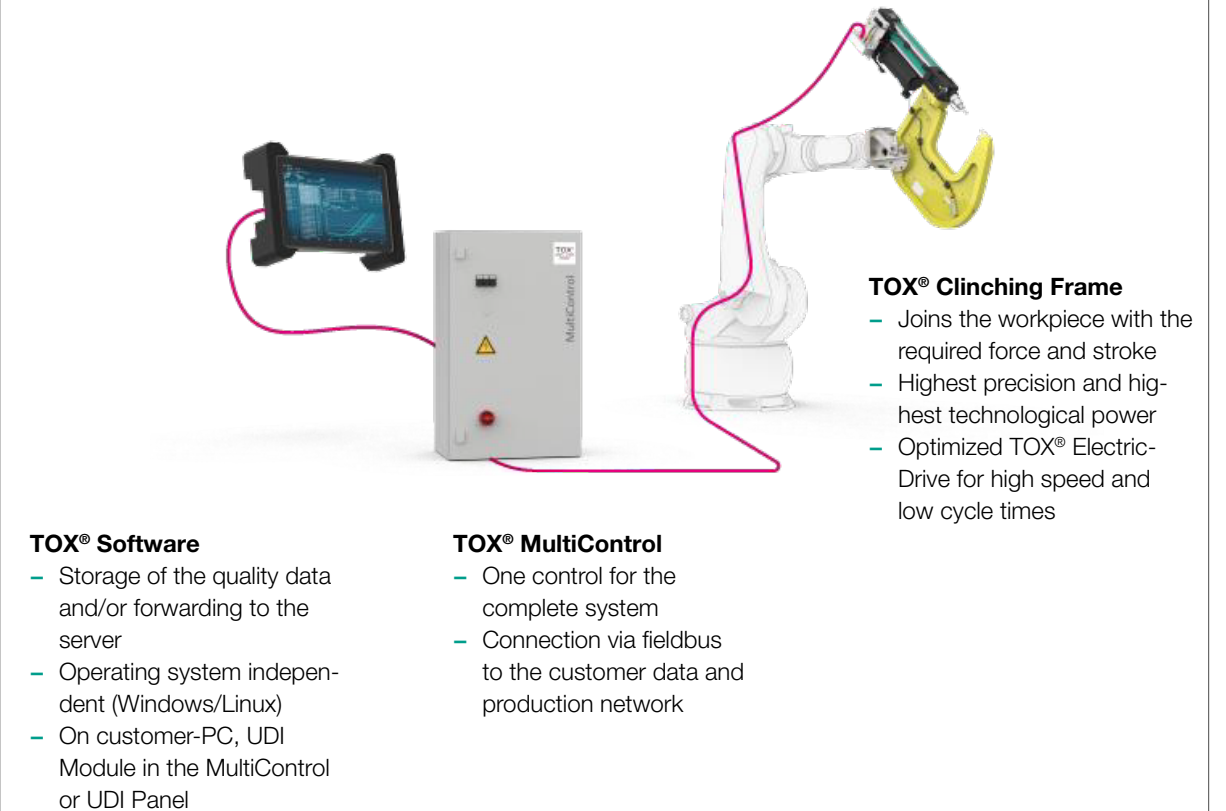
Standardized module

The clinching system consists of standardized modules. This way, we provide you with cost-optimized systems and can optimally respond to time-critical requirements

User friendliness

One control for all systems. One software for all systems. Reduce the complexity for your employees: The systems of the multi-technology platform can be comprehensively operated and maintained with just one training session.

Components of the TOX® Clinching System



TOX® Clinching System as robot guided system. Also possible as stationary system.

Options

TOX® Spraying System

For better joining results workpieces and tools can be sprayed with an emulsion e.g. parts made of stainless steel.



TOX® Die Driven - Systems

Instead of the punch, the die side moves toward the piece-part. Smart on workpieces with difficult accessibility.



TOX® Presses

Joining systems from small to large

As an experienced technology and system provider, TOX® offers compact presses, standard press systems, and special machines that are perfectly tailored to the technology.

TOX® FlexPress Compact

- Powered by TOX® ElectricPowerDrive in four variants: 5 kN, 10 kN, 30 kN or 60 kN
- Clinching, riveting, pressing in... all TOX® Technologies are feasible
- Seamless process- and quality monitoring with TOX® Software on the 15" panel
- Space-saving thanks to a compact design and integrated control cabinet
- ESD certification
- Height-adjustable table



TOX® FlexPress Station

The modular assembly workstation for joining and forming processes. Proven mechanical engineering and superior press construction meet innovative process monitoring and a sophisticated control system. From 0.05 to 200 kN.



TOX® Presses / Special machines

Thanks to its modular design, virtually any configuration is possible, from a standard press to customized special-purpose machine. Robust and economical, durable and versatile, CE- and UVV-compliant, including a comprehensive range of accessories. All technology components are integrated into a single system, including type-tested safety controls and process monitoring systems.



TOX® BatteryTong

The mobile handheld for clinching

- Press forces 10 - 50 kN
- Point diameters 5 - 12 mm
- Ca. 400 clinching points per load
- Bluetooth-interface for data transmission
- Frame sizes 30 mm or 60 mm



TOX® Clinching Tools

In joining processes, the tool set plays an important role. The TOX® Laboratory determines the appropriate tool set for each application to guarantee the holding forces of the clinch point. The tools are proven to be durable. Punches and dies are precisely matched to materials, sheet thicknesses, and joining force.



Controls and process monitoring for better production quality

TOX® Software

The TOX® Software is the comprehensive solution for the operation of your plants. In addition to pure operation, detailed visualizations and data evaluations can be viewed and processes can be configured.

The software also impresses with its modern, intuitive UI design with dashboard and widget functions. These allow easy customization of the user interface, resulting in screens that are perfectly adapted to each operator. Simply install the software on your Windows PC or use our TOX® UDI Panel. Communication takes place in real time, guaranteeing high repeat accuracy and maximum process control performance.



TOX® UDI Panel

Touchpanel panel with an integrated PC: The TOX® UDI Panel enables the control and visualisation of your system as well as the storage and processing of all process data. The powerful PC in a robust, milled aluminum housing is available in three sizes for mounting on the support arm or as a built-in version.

Perfect for monitoring and optimising your production processes. Thanks to its advanced technology, it offers you a comprehensive process monitoring solution and ensures that you always have an overview of your systems.



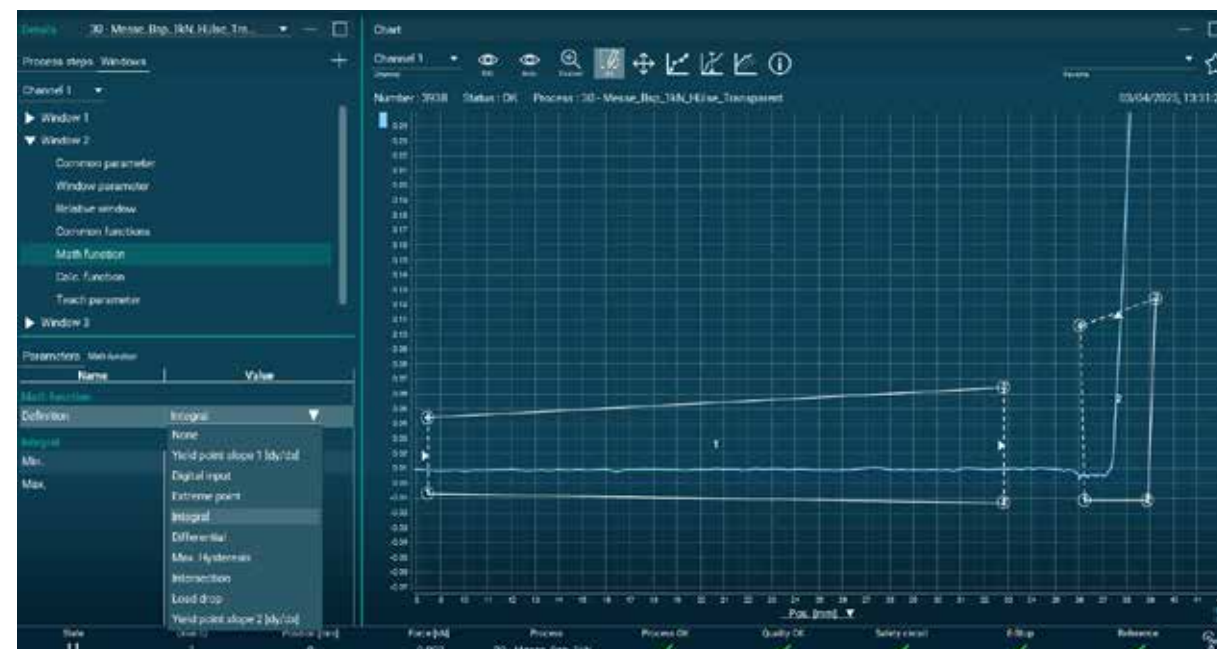
TOX® Clinching monitor CEP 600

The TOX® Clinching monitor offers precise process monitoring, in which the force values of several clinching tools can be displayed.

The CEP 600 is a powerful process monitoring system designed specifically for clinching. The system ensures continuous control and documentation of the quality parameter in real time.

Advantages

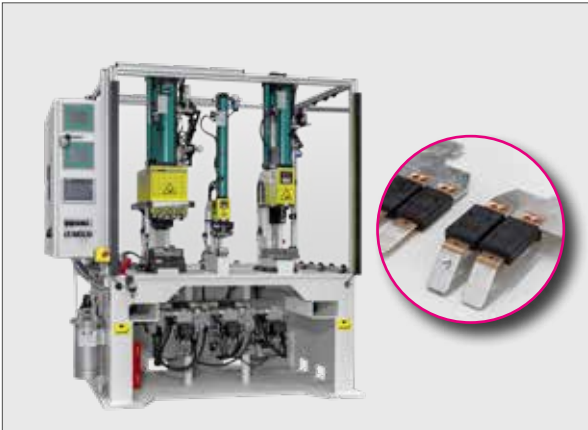
- Window technique for processes
- For manual workstations
- Built-in and wall-mounted version
- Touchscreen 7" and 10"



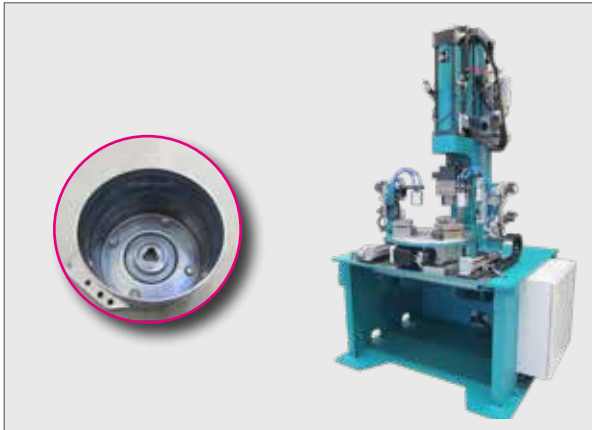
Technical Data

- For all TOX® ElectricDrive
- 5 windows per process freely definable
- 1000 software programs
- 2 channels e.g. force 1 / force 2 on position
- Flexible fieldbus with 32 words
- 10 tracks in one diagram
- 5000 diagram points per track

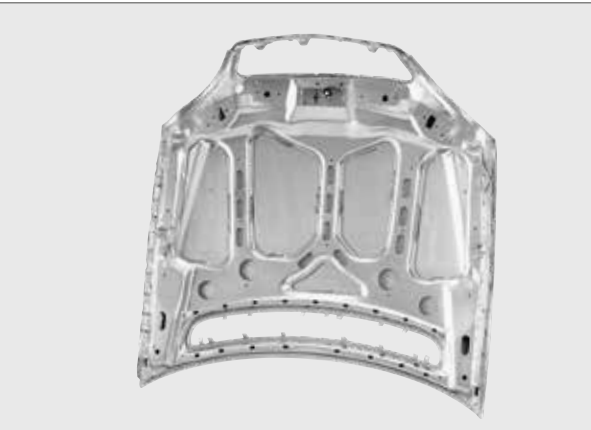
Applications in real life
Individual and flexible



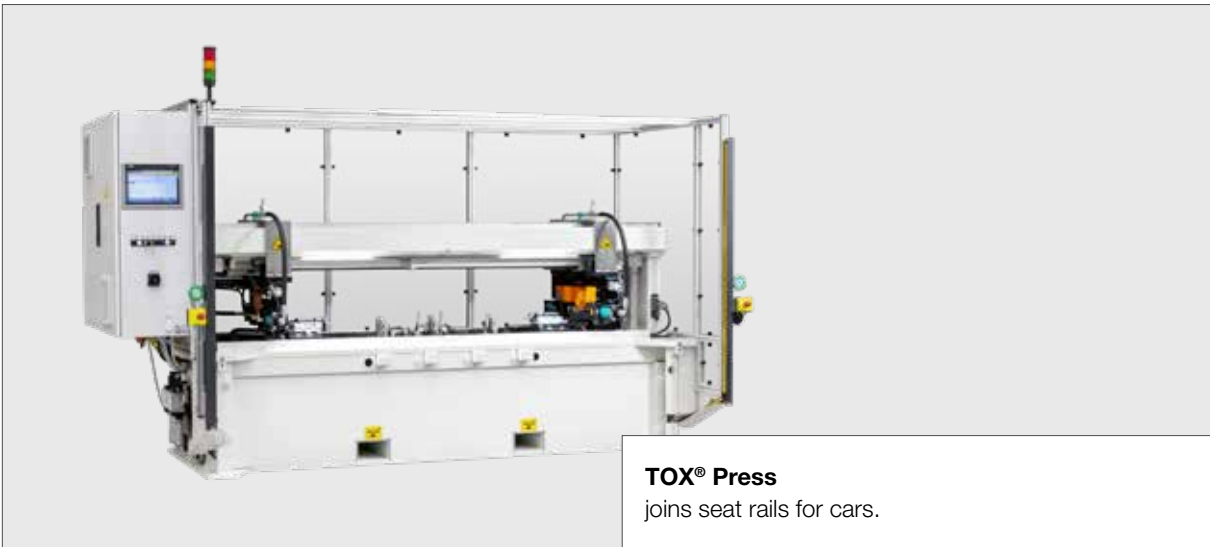
TOX® Press with TOX® Powerpackage
for eClinching of fuse-components.



TOX® Press with TOX® Powerpackage
joins housings for electric motors.



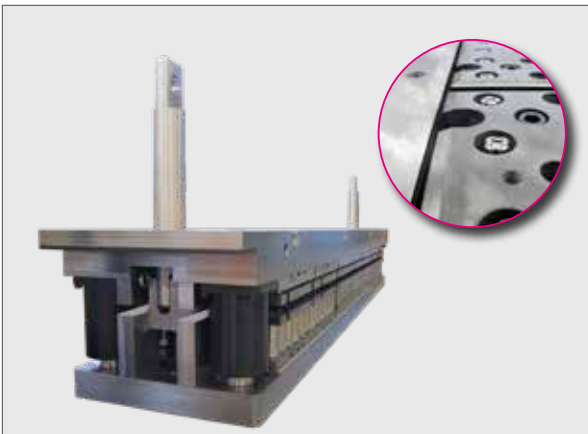
TOX® Multi Technology Platform
Clinching tongs on robots join car parts.



TOX® Press
joins seat rails for cars.



TOX® Press
joins lighting parts.



TOX® Multi-point tool
with 120 punches and dies, which is used for
joining sheet metal in an automated production.



TOX® Press
for eClinching of fuse-components.



TOX® Press
clinches refrigerator housings.

Your Partner for future-proof solutions. Globally networked. Regionally based.



Joining is tradition

For over 45 years, TOX® has been a leader in providing innovative solutions for sheet metal joining and drive technology. As an international family-owned business with its headquarters in Weingarten (Baden-Württemberg) we develop intelligent systems for joining, press-fitting, and clinching — from individual components to complex production systems.

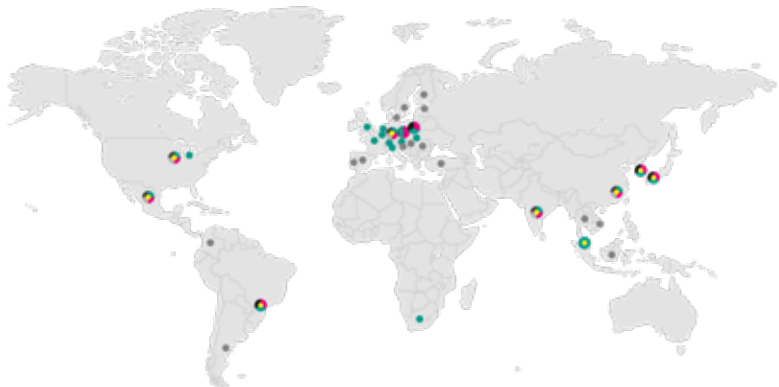
TOX® stands for the highest quality, durability and depth of integration.



Technologies are our DNA

TOX® determines the ideal technology for the respective application in advance in its in-house test laboratory. Preliminary tests are carried out and analyzed using samples and test parts.

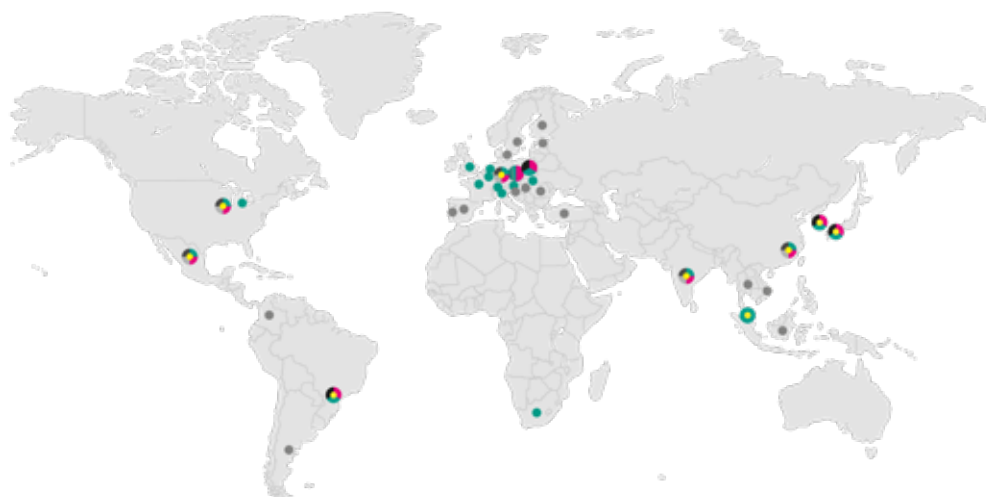
The results provide important parameters, such as the required pressing force or tool geometry, which are incorporated into the system configuration and processing technology. Final test reports verifiably ensure the quality of the joint.



German Engineering meets global Innovation

Our headquarters and technology center are located in Weingarten (Baden-Württemberg). With 17 branches worldwide and a broad network of sales partners, we offer global proximity and comprehensive service.

Five production sites – in Germany, the USA, China, India and Mexico – ensure highest quality wherever our customers are.



TOX® PRESSOTECHNIK SE & Co. KG
Riedstrasse 4
88250 Weingarten / Germany

Find your local contact partner at:
www.tox.com